gpeyt1-41 · Square footing, WT at base

GEO PE · Long Calculation Problems I — Topic 9: Shallow Foundations

Problem

A 6 ft × 6 ft column footing bears 4 ft below grade in a uniform medium-dense sand (c = 0, φ = 32°, γ = 120 pcf above and γsat ≈ 125 pcf below the water table). The seasonal-high water table is at the footing base (Dw = Df). Use the AASHTO/NHI general bearing equation with handbook shape factors and groundwater correction factors, FS = 3.0. Column dead load 160 kips, live load 90 kips (not needed to compute capacity).

Most nearly, the allowable column load (gross, FS = 3) is:

A. 295 kips
B. 256 kips
C. 199 kips
D. 148 kips

Given equations / data

> qult = cNc sc + qNq sq Cwq + 0.5γBNγ sγ Cwγ; φ = 32°: Nq = 23.2, Nγ = 30.2; sq = 1 + (B/L) tan φ; sγ = 1 − 0.4(B/L); Dw = Df → Cwq = 1.0, Cwγ = 0.5

Worked solution

1. q = 120×4 = 480 psf; sq = 1 + tan32 = 1.625; sγ = 0.6
2. Surcharge term = 480×23.2×1.625×1.0 = 18,095 psf
3. γ term = 0.5×120×6×30.2×0.6×0.5 = 3,262 psf
4. qult = 21,356 psf; qall = 7,119 psf
5. Qall = 7,119 × 36 ft² = 256 kips

Final answer

256 kips

Why the other values are wrong

- **295 kips** — ignored the groundwater correction (Cwγ = 1)
- **199 kips** — omitted shape factors
- **148 kips** — also took Cwq = 0.5 (misread the Dw = Df row)

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §3.4.2–3.4.2.1 Bearing Capacity Equation, Shape Factors, Bearing Capacity Factors and Groundwater Correction tables, pp. 90–91

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=du7B1MO-GCg, https://www.youtube.com/watch?v=ZUGdBao3tkc, https://www.youtube.com/watch?v=1fx03dLbmQc (no creator values copied).

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gpeyt1-42 · Strip footing on clay (φ = 0)

GEO PE · Long Calculation Problems I — Topic 9: Shallow Foundations

Problem

A 5 ft wide continuous wall footing bears 3.5 ft deep in a saturated, homogeneous stiff clay. Unconfined compression tests give an average qu = 2,400 psf; γ = 118 pcf; the water table is deep. Analyze the short-term (undrained) case with FS = 3.0 on net bearing pressure. The wall load is 9 kips/ft.

Most nearly, the allowable NET bearing pressure is:

A. 4,112 psf
B. 2,194 psf
C. 2,280 psf
D. 2,056 psf

Given equations / data

> φ = 0: Nc = 5.14, Nq = 1, Nγ = 0; su = qu/2; qnet,ult = qult − γDf = 5.14 su

Worked solution

1. su = 2,400/2 = 1,200 psf
2. qnet,ult = 5.14 × 1,200 = 6,168 psf
3. qnet,all = 6,168/3.0 = 2,056 psf
4. (check: wall pressure 9,000/5 = 1,800 psf gross)

Final answer

2,056 psf

Why the other values are wrong

- **4,112 psf** — used qu as su (forgot ÷2)
- **2,194 psf** — reported gross (added γDf after FS)
- **2,280 psf** — used Terzaghi Nc = 5.7

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §3.4.2 Bearing Capacity Equation (strip) and factor table (φ = 0: Nc = 5.14), pp. 90–91; §3.3.3 Undrained Shear Strength

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=l9jr9yfBzG0, https://www.youtube.com/watch?v=ZUGdBao3tkc (no creator values copied).

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gpeyt1-43 · Eccentric footing (Meyerhof)

GEO PE · Long Calculation Problems I — Topic 9: Shallow Foundations

Problem

An 8 ft × 8 ft footing supports a vertical load P = 180 kips (including footing weight) and an overturning moment M = 120 kip-ft about one axis from wind on a sign structure. The footing bears on dense sand where settlement analysis uses the Meyerhof equivalent uniform pressure. Allowable bearing pressure is 4.0 ksf.

Most nearly, the Meyerhof equivalent uniform bearing pressure is:

A. 4,219 psf
B. 2,812 psf
C. 3,375 psf
D. 3,068 psf

Given equations / data

> $e = M/P \ (\le B/6)$; $B' = B - 2e$; $q_{eq} = P/(B'L)$; $q_{max} = (P/BL)(1 + 6e/B)$

Worked solution

1. $e = 120/180 = 0.667$ ft $\le B/6 = 1.33$ ✓
2. $B' = 8 - 2(0.667) = 6.667$ ft
3. $q_{eq} = 180/(6.667 \times 8) = 3.375$ ksf $= 3{,}375$ psf
4. ($q_{max} = 4{,}219$ psf)

Final answer

3,375 psf

Why the other values are wrong

- **4,219 psf** — reported qmax (trapezoidal toe pressure)
- **2,812 psf** — ignored eccentricity
- **3,068 psf** — used $B' = B - e$

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 Applied stress at base — $q_{eq} = \Sigma V/B'$, $B' = B - 2e$ (cantilever wall figure, p. 209); NHI-06-089 §8 effective width

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=du7B1MO-GCg (no creator values copied).

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gpeyt1-44 · 2:1 stress + clay settlement

GEO PE · Long Calculation Problems I — Topic 9: Shallow Foundations

Problem

A 10 ft × 10 ft footing with net applied pressure Δp = 2,600 psf bears 3 ft below grade on sand (γ = 120 pcf above and below the water table, which is at the footing base). A 10 ft normally consolidated clay layer (γsat = 112 pcf, e0 = 0.92, Cc = 0.3) begins 10 ft below the footing base. Use the 2:1 method at mid-clay and a single layer.

Most nearly, the consolidation settlement of the clay is:

A. 9.5 in.
B. 3.6 in.
C. 2.5 in.
D. 1.5 in.

Given equations / data

> $\Delta\sigma_z = \Delta p\, B\, L/[(B + z)(L + z)]$ (2:1, z below footing base); $S = C_c H/(1 + e_0) \log(\sigma'_f/\sigma'_0)$; $\gamma_w = 62.4$ pcf

Worked solution

1. z (base → mid-clay) = 10 + 5 = 15 ft
2. $\Delta\sigma = 2{,}600 \times 10^2/(10+15)^2 = 416$ psf
3. $\sigma'_0 = 3(120) + 10(120-62.4) + 5(112-62.4) = 1{,}184$ psf
4. $S = 0.3 \times 10/1.92 \times \log(1{,}600/1{,}184) = 0.2043$ ft = 2.5 in.

Final answer

2.5 in.

Why the other values are wrong

- **9.5 in.** — applied full Δp at mid-clay (no spreading)
- **3.6 in.** — evaluated Δσ at the top of clay
- **1.5 in.** — used total stress for σ′0

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §3.5.1 Stress Distribution (2:1 method), p. 94; §3.2.1 NC consolidation, p. 83

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=NwGztxby-vQ, https://www.youtube.com/watch?v=SLtD_6vcRAs (no creator values copied).

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gpeyt1-45 · Elastic settlement

GEO PE · Long Calculation Problems I — Topic 9: Shallow Foundations

Problem

A flexible 9 ft × 9 ft mat for an equipment pad applies Δp = 3,200 psf to a deep, homogeneous dense gravelly sand with Em = 420,000 psf (≈ 2,917 psi) and ν = 0.3. Estimate the AVERAGE immediate settlement under the flexible pad (not the center).

Most nearly, the average elastic settlement is:

A. 0.71 in.
B. 0.84 in.
C. 0.78 in.
D. 0.06 in.

Given equations / data

> δv = Cd Δp Bf (1 − ν²)/Em; Cd (square, flexible): center 1.12, corner 0.56, average 0.95; square rigid 0.99

Worked solution

1. 1 − ν² = 0.91
2. δv = 0.95 × 3,200 × 9 × 0.91/420,000 = 0.0593 ft = 0.71 in.

Final answer

0.71 in.

Why the other values are wrong

- **0.84 in.** — used the center factor 1.12
- **0.78 in.** — omitted (1 − ν²)
- **0.06 in.** — left the answer in ft labeled in.

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §3.5.2 Settlement (Elastic Method) and Cd table, p. 95

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=SLtD_6vcRAs (no creator values copied).

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